

18

Integration in the Complex Plane

EXERCISES 18.1

Contour Integrals

- $\int_C (z + 3) dz = (2 + 4i) \left[\int_1^3 (2t + 3) dt + i \int_1^3 (4t - 1) dt \right] = (2 + 4i)[14 + 14i] = -28 + 84i$
- $\int_C (2\bar{z} - z) dz = \int_0^2 [-t - 3(t^2 + 2)i](-1 + 2ti) dt = \int_0^2 (6t^3 + 13t) dt + i \int_0^2 (t^2 + 2) dt = 50 + \frac{20}{3}i$
- $\int_C z^2 dz = (3 + 2i)^3 \int_{-2}^2 t^2 dt = \frac{16}{3}(3 + 2i)^3 = -48 + \frac{736}{3}i$
- $\int_C (3z^2 - 2z) dz = \int_0^1 (-15t^4 + 4t^3 + 3t^2 - 2t) dt + i \int_0^1 (-6t^5 + 12t^3 - 6t^2) dt = -2 + 0i = -2$
- Using $z = e^{it}$, $-\pi/2 \leq t \leq \pi/2$, and $dz = ie^{it} dt$, $\int_C \frac{1+z}{z} dz = - \int_{-\pi/2}^{\pi/2} (1 + e^{it}) dt = (2 + \pi)i$.
- $\int_C |z|^2 dz = \int_1^2 \left(2t^5 + \frac{2}{t} \right) dt - i \int_1^2 \left(t^2 + \frac{1}{t^4} \right) dt = 21 + \ln 4 - \frac{21}{8}i$
- Using $z = e^{it} = \cos t + i \sin t$, $dz = (-\sin t + i \cos t) dt$ and $x = \cos t$,

$$\begin{aligned} \oint_C \operatorname{Re}(z) dz &= \int_0^{2\pi} \cos t (-\sin t + i \cos t) dt = - \int_0^{2\pi} \sin t \cos t dt + i \int_0^{2\pi} \cos^2 t dt \\ &= -\frac{1}{2} \int_0^{2\pi} \sin 2t dt + \frac{1}{2} i \int_0^{2\pi} (1 + \cos 2t) dt = \pi i. \end{aligned}$$

- Using $z + i = e^{it}$, $0 \leq t \leq 2\pi$, and $dz = ie^{it} dt$,

$$\oint_C \left[\frac{1}{(z+i)^3} - \frac{5}{z+i} + 8 \right] dz = i \int_0^{2\pi} [e^{-2it} - 5 + 8e^{it}] dt = -10\pi i.$$

- Using $y = -x + 1$, $0 \leq x \leq 1$, $z = x + (-x + 1)i$, $dz = (1 - i) dx$,

$$\int_C (x^2 + iy^3) dz = (1 - i) \int_1^0 [x^2 + (1 - x)^3 i] dx = -\frac{7}{12} + \frac{1}{12}i.$$

- Using $z = e^{it}$, $\pi \leq t \leq 2\pi$, $dz = ie^{it} dt$, $x = \cos t = (e^{it} + e^{-it})/2$, $y = \sin t = (e^{it} - e^{-it})/2i$,

$$\begin{aligned} \int_C (x^3 - iy^3) dz &= \frac{1}{8} i \int_{\pi}^{2\pi} (e^{3it} + 3e^{it} + 3e^{-it} + e^{-3it}) e^{it} dt + \frac{1}{8} i \int_{\pi}^{2\pi} (e^{3it} - 3e^{it} + 3e^{-it} - e^{-3it}) e^{it} dt \\ &= \frac{1}{8} i \int_{\pi}^{2\pi} (2e^{4it} + 6) dt = \frac{3\pi}{4} i. \end{aligned}$$

18.1 Contour Integrals

11. $\int_C e^z dz = \int_{C_1} e^z dz + \int_{C_2} e^z dz$ where C_1 and C_2 are the line segments $y = 0, 0 \leq x \leq 2$ and $y = -\pi x + 2\pi, 1 \leq x \leq 2$, respectively. Now

$$\int_{C_1} e^z dz = \int_0^2 e^x dx = e^2 - 1$$

$$\int_{C_2} e^z dz = (1 - \pi i) \int_2^1 e^{x + (-\pi x + 2\pi)i} dx = (1 - \pi i) \int_2^1 e^{(1-\pi i)x} dx = e^{1-\pi i} - e^{2(1-\pi i)} = -e - e^2.$$

In the second integral we have used the fact that e^z has period $2\pi i$. Thus

$$\int_C e^z dz = (e^2 - 1) + (-e - e^2) = -1 - e.$$

12. $\int_C \sin z dz = \int_{C_1} \sin z dz + \int_{C_2} \sin z dz$ where C_1 and C_2 are the line segments $y = 0, 0 \leq x \leq 1$, and $x = 1, 0 \leq y \leq 1$, respectively. Now

$$\int_{C_1} \sin z dz = \int_0^1 \sin x dx = 1 - \cos 1$$

$$\int_{C_2} \sin z dz = i \int_0^1 \sin(1 + iy) dy = \cos 1 - \cos(1 + i).$$

Thus

$$\int_C \sin z dz = (1 - \cos 1) + (\cos 1 - \cos(1 + i)) = 1 - \cos(1 + i) = (1 - \cos 1 \cosh 1) + i \sin 1 \sinh 1 = 0.1663 + 0.9889i.$$

13. We have $\int_C \operatorname{Im}(z - i) dz = \int_{C_1} (y - 1) dz + \int_{C_2} (y - 1) dz$

On $C_1, z = e^{it}, 0 \leq t \leq \pi/2, dz = ie^{it} dt, y = \sin t = (e^{it} - e^{-it})/2i,$

$$\int_{C_1} (y - 1) dz = \frac{1}{2} \int_0^{\pi/2} [e^{it} - e^{-it} - 2i] e^{it} dt = \frac{1}{2} \int_0^{\pi/2} [e^{2it} - 1 + 2ie^{it}] dt = 1 - \frac{\pi}{4} - \frac{1}{2}i.$$

On $C_2, y = x + 1, -1 \leq x \leq 0, z = x + (x + 1)i, dz = (1 + i) dx,$

$$\int_{C_2} (y - 1) dz = (1 + i) \int_0^{-1} x dx = \frac{1}{2} + \frac{1}{2}i.$$

Thus

$$\int_C \operatorname{Im}(z - i) dz = \left(1 - \frac{\pi}{4} - \frac{1}{2}i\right) + \left(\frac{1}{2} + \frac{1}{2}i\right) = \frac{3}{2} - \frac{\pi}{4}.$$

14. Using $x = 6 \cos t, y = 2 \sin t, \pi/2 \leq t \leq 3\pi/2, z = 6 \cos t + 2i \sin t, dz = (-6 \sin t + 2i \cos t) dt,$

$$\int_C dz = -6 \int_{\pi/2}^{3\pi/2} \sin t dt + 2i \int_{\pi/2}^{3\pi/2} \cos t dt = 2i(-2) = -4i.$$

15. We have $\oint_C ze^z dz = \int_{C_1} ze^z dz + \int_{C_2} ze^z dz + \int_{C_3} ze^z dz + \int_{C_4} ze^z dz$

On $C_1, y = 0, 0 \leq x \leq 1, z = x, dz = dx,$

$$\int_{C_1} ze^z dz = \int_0^1 xe^x dx = xe^x - e^x \Big|_0^1 = 1.$$

On C_2 , $x = 1$, $0 \leq y \leq 1$, $z = 1 + iy$, $dz = i dy$,

$$\int_{C_2} ze^z dz = i \int_0^1 (1 + iy)e^{1+iy} dy = ie^{i+1}.$$

On C_3 , $y = 1$, $0 \leq x \leq 1$, $z = x + i$, $dz = dx$,

$$\int_{C_3} ze^z dz = \int_1^0 (x + i)e^{x+i} dx = (i - 1)e^i - ie^{1+i}.$$

On C_4 , $x = 0$, $0 \leq y \leq 1$, $z = iy$, $dz = i dy$,

$$\int_{C_4} ze^z dz = - \int_1^0 ye^{iy} dy = (1 - i)e^i - 1.$$

Thus

$$\oint_C ze^z dz = 1 + ie^{i+1} + (i - 1)e^i - ie^{1+i} + (1 - i)e^i - 1 = 0.$$

16. We have

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz$$

On C_1 , $y = x^2$, $-1 \leq x \leq 0$, $z = x + ix^2$, $dz = (1 + 2xi) dx$,

$$\int_{C_1} f(z) dz = \int_{-1}^0 2(1 + 2xi) dx = 2 - 2i.$$

On C_2 , $y = x^2$, $0 \leq x \leq 1$, $z = x + ix^2$, $dz = (1 + 2xi) dx$,

$$\int_{C_2} f(z) dz = \int_0^1 6x(1 + 2xi) dx = 3 + 4i.$$

Thus

$$\int_C f(z) dz = 2 - 2i + 3 + 4i = 5 + 2i.$$

17. We have

$$\oint_C x dz = \int_{C_1} x dz + \int_{C_2} x dz + \int_{C_3} x dz$$

On C_1 , $y = 0$, $0 \leq x \leq 1$, $z = x$, $dz = dx$,

$$\int_{C_1} x dz = \int_0^1 x dx = \frac{1}{2}.$$

On C_2 , $x = 1$, $0 \leq y \leq 1$, $z = 1 + iy$, $dz = i dy$,

$$\int_{C_2} x dz = i \int_0^1 dy = i.$$

On C_3 , $y = x$, $0 \leq x \leq 1$, $z = x + ix$, $dz = (1 + i) dx$,

$$\int_{C_3} x dz = (1 + i) \int_1^0 x dx = -\frac{1}{2} - \frac{1}{2}i.$$

Thus

$$\oint_C x dz = \frac{1}{2} + i - \frac{1}{2} - \frac{1}{2}i = \frac{1}{2}i.$$

18. We have

$$\oint_C (2z - 1) dz = \int_{C_1} (2z - 1) dz + \int_{C_2} (2z - 1) dz + \int_{C_3} (2z - 1) dz$$

On C_1 , $y = 0$, $0 \leq x \leq 1$, $z = x$, $dz = dx$,

$$\int_{C_1} (2z - 1) dz = \int_0^1 (2x - 1) dx = 0.$$

18.1 Contour Integrals

On C_2 , $x = 1$, $0 \leq y \leq 1$, $z = 1 + iy$, $dz = i dy$,

$$\int_{C_2} (2z - 1) dz = -2 \int_0^1 y dy + i \int_0^1 dy = -1 + i.$$

On C_3 , $y = x$, $z = x + ix$, $dz = (1 + i) dx$,

$$\int_{C_3} (2z - 1) dz = (1 + i) \int_1^0 (2x - 1 + 2ix) dx = 1 - i.$$

Thus

$$\oint_C (2z - 1) dz = 0 - 1 + i + 1 - i = 0.$$

19. We have

$$\oint_C z^2 dz = \int_{C_1} z^2 dz + \int_{C_2} z^2 dz + \int_{C_3} z^2 dz$$

On C_1 $y = 0$, $0 \leq x \leq 1$, $z = x$, $dz = dx$,

$$\int_{C_1} z^2 dz = \int_0^1 x^2 dx = \frac{1}{3}.$$

On C_2 , $x = 1$, $0 \leq y \leq 1$, $z = 1 + iy$, $dz = i dy$,

$$\int_{C_2} z^2 dz = \int_0^1 (1 + iy)^2 i dy = -1 + \frac{2}{3} i.$$

On C_3 , $y = x$, $0 \leq x \leq 1$, $z = x + ix$, $dz = (1 + i) dx$,

$$\int_{C_3} z^2 dz = (1 + i)^3 \int_1^0 x^2 dx = \frac{2}{3} - \frac{2}{3} i.$$

Thus

$$\oint_C z^2 dz = \frac{1}{3} - 1 + \frac{2}{3} i + \frac{2}{3} - \frac{2}{3} i = 0.$$

20. We have

$$\oint_C \bar{z}^2 dz = \int_{C_1} \bar{z}^2 dz + \int_{C_2} \bar{z}^2 dz + \int_{C_3} \bar{z}^2 dz$$

On C_1 , $y = 0$, $0 \leq x \leq 1$, $z = x$, $dz = dx$,

$$\int_{C_1} \bar{z}^2 dz = \int_0^1 x^2 dx = \frac{1}{3}.$$

On C_2 , $x = 1$, $0 \leq y \leq 1$, $z = 1 + iy$, $dz = i dy$,

$$\int_{C_2} \bar{z}^2 dz = - \int_0^1 (1 - iy)^2 (-i dy) = 1 + \frac{2}{3} i.$$

On C_3 , $y = x$, $0 \leq x \leq 1$, $z = x + ix$, $dz = (1 + i) dx$,

$$\int_{C_3} \bar{z}^2 dz = (1 - i)^2 (1 + i) \int_1^0 x^2 dx = -\frac{2}{3} + \frac{2}{3} i.$$

Thus

$$\oint_C \bar{z}^2 dz = \frac{1}{3} + 1 + \frac{2}{3} i - \frac{2}{3} + \frac{2}{3} i = \frac{2}{3} + \frac{4}{3} i.$$

21. On C , $y = -x + 1$, $0 \leq x \leq 1$, $z = x + (-x + 1)i$, $dz = (1 - i) dx$,

$$\int_C (z^2 - z + 2) dz = (1 - i) \int_0^1 [x^2 - (1 - x)^2 - x + 2 + (3x - 2x^2 - 1)i] dx = \frac{4}{3} - \frac{5}{3} i.$$

22. We have

$$\int_C (z^2 - z + 2) dz = \int_{C_1} (z^2 - z + 2) dz + \int_{C_2} (z^2 - z + 2) dz$$

On C_1 , $y = 1$, $0 \leq x \leq 1$, $z = x + i$, $dz = dx$,

$$\int_{C_1} (z^2 - z + 2) dz = \int_0^1 [(x + i)^2 - x + 2 - i] dx = \frac{5}{6}.$$

On C_2 , $x = 1$, $0 \leq y \leq 1$, $z = 1 + iy$, $dz = i dy$,

$$\int_{C_2} (z^2 - z + 2) dz = i \int_1^0 [(1 + iy)^2 + 1 - iy] dy = \frac{1}{2} - \frac{5}{3}i.$$

Thus

$$\int_C (z^2 - z + 2) dz = \frac{1}{2} - \frac{5}{3}i + \frac{5}{6} = \frac{4}{3} - \frac{5}{3}i.$$

23. On C , $y = 1 - x^2$, $0 \leq x \leq 1$, $z = x + i(1 - x^2)$, $dz = (1 - 2xi) dx$,

$$\int_C (z^2 - z + 2) dz = \int_0^1 (-5x^4 + 2x^3 + 7x^2 - 3x + 1) dx + i \int_0^1 (2x^5 - 8x^3 + 3x^2 - 1) dx = \frac{4}{3} - \frac{5}{3}i.$$

24. On C , $x = \sin t$, $y = \cos t$, $0 \leq t \leq \pi/2$ or $z = ie^{-it}$, $dz = e^{-it} dt$,

$$\begin{aligned} \int_C (z^2 - z + 2) dz &= \int_0^{\pi/2} (-e^{-2it} - ie^{-it} + 2)e^{-it} dt = \int_0^{\pi/2} (-e^{-3it} - ie^{-2it} + 2e^{-it}) dt \\ &= -\frac{1}{3}ie^{-3\pi i/2} + \frac{1}{2}e^{-\pi i} + 2ie^{-\pi i/2} + \frac{1}{3}i - \frac{1}{2} - 2i = \frac{4}{3} - \frac{5}{3}i. \end{aligned}$$

25. On C , $\left| \frac{e^z}{z^2 + 1} \right| \leq \frac{|e^z|}{|z|^2 - 1} = \frac{e^5}{24}$. Thus $\left| \oint_C \frac{e^z}{z^2 + 1} dz \right| \leq \frac{e^5}{24} \cdot 10\pi = \frac{5\pi}{12} e^5$.

26. On C , $\left| \frac{1}{z^2 - 2i} \right| \leq \frac{1}{|z|^2 - |2i|} = \frac{1}{34}$. Thus $\left| \int_C \frac{1}{z^2 - 2i} dz \right| \leq \frac{1}{34} \cdot \frac{1}{2}(12\pi) = \frac{3\pi}{17}$.

27. The length of the line segment from $z = 0$ to $z = 1 + i$ is $\sqrt{2}$. In addition, on this line segment

$$|z^2 + 4| \leq |z|^2 + 4 \leq |1 + i|^2 + 4 = 6.$$

$$\text{Thus } \left| \int_C (z^2 + 4) dz \right| \leq 6\sqrt{2}.$$

28. On C , $\left| \frac{1}{z^3} \right| = \frac{1}{|z|^3} = \frac{1}{64}$. Thus $\left| \int_C \frac{1}{z^3} dz \right| \leq \frac{1}{64} \cdot \frac{1}{4}(8\pi) = \frac{\pi}{32}$.

$$\begin{aligned} 29. \text{ (a) } \int_C dz &= \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \Delta z_k = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (z_k - z_{k-1}) \\ &= \lim_{\|P\| \rightarrow 0} [(z_1 - z_0) + (z_2 - z_1) + (z_3 - z_2) + \cdots + (z_{n-1} - z_{n-2}) + (z_n - z_{n-1})] \\ &= \lim_{\|P\| \rightarrow 0} (z_n - z_0) = z_n - z_0 \end{aligned}$$

(b) With $z_n = -2i$ and $z_0 = 2i$, $\int_C dz = -2i - (2i) = -4i$.

30. With $z_k^* = z_k$,

$$\begin{aligned} \int_C z dz &= \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n z_k (z_k - z_{k-1}) \\ &= \lim_{\|P\| \rightarrow 0} [(z_1^2 - z_1 z_0) + (z_2^2 - z_2 z_1) + \cdots + (z_n^2 - z_n z_{n-1})]. \end{aligned} \quad (1)$$

18.1 Contour Integrals

With $z_k^* = z_{k-1}$,

$$\begin{aligned}\int_C z dz &= \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n z_{k-1}(z_k - z_{k-1}) \\ &= \lim_{\|P\| \rightarrow 0} [(z_0 z_1 - z_0^2) + (z_1 z_2 - z_1^2) + \cdots + (z_{n-1} z_n - z_{n-1}^2)].\end{aligned}\quad (2)$$

Adding (1) and (2) gives

$$2 \int_C z dz = \lim_{\|P\| \rightarrow 0} (z_n^2 - z_0^2) \quad \text{or} \quad \int_C z dz = \frac{1}{2}(z_n^2 - z_0^2).$$

$$31. \quad (a) \quad \int_C (6z + 4) dz = 6 \int_C z dz + 4 \int_C dz = \frac{6}{2}[(2 + 3i)^2 - (1 + i)^2] + 4[(2 + 3i) - (1 + i)] = -11 + 38i$$

(b) Since the contour is closed, $z_0 = z_n$ and so

$$6 \int_C z dz + 4 \int_C dz = 6[z_0^2 - z_0^2] + 4[z_0 - z_0] = 0.$$

32. For $f(z) = 1/z$, $\overline{f(z)} = 1/\bar{z}$, so on $z = 2e^{it}$, $\bar{z} = 2e^{-it}$, $dz = 2ie^{it} dt$, and

$$\oint_C \overline{f(z)} dz = \int_0^{2\pi} \frac{1}{2e^{-it}} \cdot 2ie^{it} dt = \frac{1}{2}e^{2it} \Big|_0^{2\pi} = \frac{1}{2}[e^{4\pi i} - 1] = 0.$$

Thus circulation = $\operatorname{Re} \left(\oint_C \overline{f(z)} dz \right) = 0$, and net flux = $\operatorname{Im} \left(\oint_C \overline{f(z)} dz \right) = 0$.

33. For $f(z) = 2z$, $\overline{f(z)} = 2\bar{z}$, so on $z = e^{it}$, $\bar{z} = e^{-it}$, $dz = ie^{it} dt$, and

$$\oint_C \overline{f(z)} dz = \int_0^{2\pi} (e^{-it})(ie^{it} dt) = 2i \int_0^{2\pi} dt = 4\pi i.$$

Thus circulation = $\operatorname{Re} \left(\oint_C \overline{f(z)} dz \right) = 0$, and net flux = $\operatorname{Im} \left(\oint_C \overline{f(z)} dz \right) = 4\pi$.

34. For $f(z) = 1/(z-1)$, $\overline{f(z)} = 1/(\bar{z}-1)$, so on $z-1 = 2e^{it}$, $dz = 2ie^{it} dt$, and

$$\oint_C \overline{f(z)} dz = \int_0^{2\pi} \frac{1}{2e^{it}} \cdot 2ie^{it} dt = i \int_0^{2\pi} dt = 2\pi i.$$

Thus circulation = $\operatorname{Re} \left(\oint_C \overline{f(z)} dz \right) = 0$, and net flux = $\operatorname{Im} \left(\oint_C \overline{f(z)} dz \right) = 2\pi$.

35. For $f(z) = \bar{z}$, $\overline{f(z)} = z$ so on the square we have

$$\oint_C \overline{f(z)} dz = \int_{C_1} z dz + \int_{C_2} z dz + \int_{C_3} z dz + \int_{C_4} z dz$$

where C_1 is $y = 0$, $0 \leq x \leq 1$, C_2 is $x = 1$, $0 \leq y \leq 1$, C_3 is $y = 1$, $0 \leq x \leq 1$, and C_4 is $x = 0$, $0 \leq y \leq 1$. Thus

$$\begin{aligned}\int_{C_1} z dz &= \int_0^1 x dx = \frac{1}{2} \\ \int_{C_2} z dz &= i \int_0^1 (1 + iy) dy = -\frac{1}{2} + i \\ \int_{C_3} z dz &= \int_1^0 (x + i) dx = -\frac{1}{2} - i \\ \int_{C_4} z dz &= -\int_1^0 y dy = \frac{1}{2}\end{aligned}$$

and so

$$\oint_C \overline{f(z)} dz = \frac{1}{2} + \left(-\frac{1}{2} + i\right) + \left(-\frac{1}{2} - i\right) + \frac{1}{2} = 0$$

$$\text{circulation} = \operatorname{Re} \left(\oint_C \overline{f(z)} dz \right) = \operatorname{Re}(0) = 0$$

$$\text{net flux} = \operatorname{Im} \left(\oint_C \overline{f(z)} dz \right) = \operatorname{Im}(0) = 0.$$

EXERCISES 18.2

Cauchy-Goursat Theorem

1. $f(z) = z^3 - 1 + 3i$ is a polynomial and so is an entire function.
2. z^2 is entire and $\frac{1}{z-4}$ is analytic within and on the circle $|z| = 1$.
3. $f(z) = \frac{z}{2z+3}$ is discontinuous at $z = -3/2$ but is analytic within and on the circle $|z| = 1$.
4. $f(z) = \frac{z-3}{z^2+2z+2}$ is discontinuous at $z = -1+i$ and at $z = -1-i$ but is analytic within and on the circle $|z| = 1$.
5. $f(z) = \frac{\sin z}{(z^2-25)(z^2+9)}$ is discontinuous at $z = \pm 5$ and at $z = \pm 3i$ but is analytic within and on the circle $|z| = 1$.
6. $f(z) = \frac{e^z}{2z^2+11z+15}$ is discontinuous at $z = -5/2$ and at $z = -3$ but is analytic within and on the circle $|z| = 1$.
7. $f(z) = \tan z$ is discontinuous at $z = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$ but is analytic within and on the circle $|z| = 1$.
8. $f(z) = \frac{z^2-9}{\cosh z}$ is discontinuous at $\frac{\pi}{2}i, \pm \frac{3\pi}{2}i, \dots$ but is analytic within and on the circle $|z| = 1$.
9. By the principle of deformation of contours we can choose the more convenient circular contour C_1 defined by $|z| = 1$. Thus

$$\oint_C \frac{1}{z} dz = \oint_{C_1} \frac{1}{z} dz = 2\pi i$$

by (4) of Section 18.2.

10. By the principle of deformation of contours we can choose the more convenient circular contour C_1 defined by $|z - (-1-i)| = \frac{1}{16}$. Thus

$$\oint_C \frac{5}{z+1+i} dz = 5 \oint_{C_1} \frac{1}{z - (-1-i)} dz = 5(2\pi i) = 10\pi i$$

by (4) of Section 18.2.

18.2 Cauchy-Goursat Theorem

11. By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \left(z + \frac{1}{z} \right) dz = \oint_C z dz + \oint_C \frac{1}{z} dz = 0 + 2\pi i = 2\pi i.$$

12. By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \left(z + \frac{1}{z^2} \right) dz = \oint_C \frac{1}{z} dz + \oint_C \frac{1}{z^2} dz = 0 + 0 = 0.$$

13. Since $f(z) = \frac{z}{z^2 - \pi^2}$ is analytic within and on C it follows from Theorem 18.4 that $\oint_C \frac{z}{z^2 - \pi^2} dz = 0$.

14. By (4) of Section 18.2, $\oint_C \frac{10}{(z+i)^4} dz = 0$.

15. By partial fractions, $\oint_C \frac{2z+1}{z(z+1)} dz = \oint_C \frac{1}{z} dz + \oint_C \frac{1}{z+1} dz$.

(a) By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{1}{z} dz + \oint_C \frac{1}{z+1} dz = 2\pi i + 0 = 2\pi i.$$

(b) By writing $\oint_C = \oint_{C_1} + \oint_{C_2}$ where C_1 and C_2 are the circles $|z| = 1/2$ and $|z+1| = 1/2$, respectively,

we have by Theorem 18.4 and (4) of Section 18.2,

$$\begin{aligned} \oint_C \frac{1}{z} dz + \oint_C \frac{1}{z+1} dz &= \oint_{C_1} \frac{1}{z} dz + \oint_{C_1} \frac{1}{z+1} dz + \oint_{C_2} \frac{1}{z} dz + \oint_{C_2} \frac{1}{z+1} dz \\ &= 2\pi i + 0 + 0 + 2\pi i = 4\pi i. \end{aligned}$$

(c) Since $f(z) = \frac{2z+1}{z(z+1)}$ is analytic within and on C it follows from Theorem 18.4 that

$$\oint_C \frac{2z+1}{z^2+z} dz = 0.$$

16. By partial fractions, $\oint_C \frac{2z}{z^2+3} dz = \oint_C \frac{1}{z+\sqrt{3}i} dz + \oint_C \frac{1}{z-\sqrt{3}i} dz$.

(a) By Theorem 18.4,

$$\oint_C \frac{1}{z+\sqrt{3}i} dz + \oint_C \frac{1}{z-\sqrt{3}i} dz = 0 + 0 = 0.$$

(b) By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{1}{z+\sqrt{3}i} dz + \oint_C \frac{1}{z-\sqrt{3}i} dz = 0 + 2\pi i = 2\pi i.$$

(c) By writing $\oint_C = \oint_{C_1} + \oint_{C_2}$ where C_1 and C_2 are the circles $|z+\sqrt{3}i| = 1/2$ and $|z-\sqrt{3}i| = 1/2$,

respectively, we have by Theorem 18.4 and (4) of Section 18.2,

$$\begin{aligned} \oint_C \frac{1}{z+\sqrt{3}i} dz + \oint_C \frac{1}{z-\sqrt{3}i} dz &= \oint_{C_1} \frac{1}{z+\sqrt{3}i} dz + \oint_{C_1} \frac{1}{z-\sqrt{3}i} dz + \oint_{C_2} \frac{1}{z+\sqrt{3}i} dz + \oint_{C_2} \frac{1}{z-\sqrt{3}i} dz \\ &= 2\pi i + 0 + 0 + 2\pi i = 4\pi i. \end{aligned}$$

17. By partial fractions, $\oint_C \frac{-3z+2}{z^2-8z+12} dz = \oint_C \frac{1}{z-2} dz - 4 \oint_C \frac{1}{z-6} dz$.

(a) By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{1}{z-2} dz - 4 \oint_C \frac{1}{z-6} dz = 0 - 4(2\pi i) = -8\pi i.$$

(b) By writing $\oint_C = \oint_{C_1} + \oint_{C_2}$ where C_1 and C_2 are the circles $|z-2|=1$ and $|z-6|=1$, respectively,

we have by Theorem 18.4 and (4) of Section 18.2,

$$\begin{aligned} \oint_C \frac{1}{z-2} dz - 4 \oint_C \frac{1}{z-6} dz &= \oint_{C_1} \frac{1}{z-2} dz - 4 \oint_{C_1} \frac{1}{z-6} dz + \oint_{C_2} \frac{1}{z-2} dz - 4 \oint_{C_2} \frac{1}{z-6} dz \\ &= 2\pi i - 4(0) + 0 - 4(2\pi i) = -6\pi i. \end{aligned}$$

18. (a) By writing $\oint_C = \oint_{C_1} + \oint_{C_2}$ where C_1 and C_2 are the circles $|z+2|=1$ and $|z-2i|=1$, respectively, we have by Theorem 18.4 and (4) of Section 18.2,

$$\begin{aligned} \oint_C \left(\frac{3}{z+2} - \frac{1}{z-2i} \right) dz &= \oint_{C_1} \frac{3}{z+2} dz - \oint_{C_1} \frac{1}{z-2i} dz + \oint_{C_2} \frac{3}{z+2} dz - \oint_{C_2} \frac{1}{z-2i} dz \\ &= 3(2\pi i) - 0 + 0 - 2\pi i = 4\pi i. \end{aligned}$$

19. By partial fractions,

$$\oint_C \frac{z-1}{z(z-i)(z-3i)} dz = \frac{1}{3} \oint_C \frac{1}{z} dz + \left(-\frac{1}{2} + \frac{1}{2}i \right) \oint_C \frac{1}{z-i} dz + \left(\frac{1}{6} - \frac{1}{2}i \right) \oint_C \frac{1}{z-3i} dz.$$

By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{z-1}{z(z-i)(z-3i)} dz = 0 + \left(-\frac{1}{2} + \frac{1}{2}i \right) 2\pi i + 0 = \pi(-1-i).$$

20. By partial fractions,

$$\oint_C \frac{1}{z^3+2iz^2} dz = \frac{1}{4} \oint_C \frac{1}{z} dz - \frac{1}{2}i \oint_C \frac{1}{z^2} dz - \frac{1}{4} \oint_C \frac{1}{z+2i} dz.$$

By Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{1}{z^3+2iz^2} dz = \frac{1}{4} 2\pi i - \frac{1}{2}i(0) - \frac{1}{4}(0) = \frac{\pi}{2}i.$$

21. We have

$$\oint_C \frac{8z-3}{z^2-z} dz = \oint_{C_1} \frac{8z-3}{z^2-z} dz - \oint_{C_2} \frac{8z-3}{z^2-z} dz$$

where C_1 and C_2 are the closed portions of the curve C enclosing $z=0$ and $z=1$, respectively. By partial fractions, Theorem 18.4, and (4) of Section 18.2,

$$\begin{aligned} \oint_{C_1} \frac{8z-3}{z^2-z} dz &= 5 \oint_{C_1} \frac{1}{z-1} dz + 3 \oint_{C_1} \frac{1}{z} dz = 5(0) + 3(2\pi i) = 6\pi i \\ \oint_{C_2} \frac{8z-3}{z^2-z} dz &= 5 \oint_{C_2} \frac{1}{z-1} dz + 3 \oint_{C_2} \frac{1}{z} dz = 5(2\pi i) + 3(0) = 10\pi i. \end{aligned}$$

Thus

$$\oint_C \frac{8z-3}{z^2-z} dz = 6\pi i - 10\pi i = -4\pi i.$$

18.2 Cauchy-Goursat Theorem

22. By choosing the more convenient contour C_1 defined by $|z - z_0| = r$ where r is small enough so that the circle C_1 lies entirely within C we can write

$$\oint_C \frac{1}{(z - z_0)^n} dz = \oint_{C_1} \frac{1}{(z - z_0)^n} dz.$$

Let $z - z_0 = re^{it}$, $0 \leq t \leq 2\pi$ and $dz = ire^{it} dt$. Then for $n = 1$:

$$\oint_{C_1} \frac{1}{z - z_0} dz = \int_0^{2\pi} \frac{1}{re^{it}} ire^{it} dt = i \int_0^{2\pi} dt = 2\pi i.$$

For $n \neq 1$:

$$\oint_{C_1} \frac{1}{(z - z_0)^n} dz = \frac{i}{r^{n-1}} \int_0^{2\pi} e^{(1-n)it} dt = \frac{i}{r^{n-1}} \frac{e^{(1-n)it}}{i(1-n)} \Big|_0^{2\pi} = \frac{1}{r^{n-1}(1-n)} [e^{2\pi(1-n)i} - 1] = 0$$

since $e^{2\pi(1-n)i} = 1$.

23. Write
$$\oint_C \left(\frac{e^z}{z+3} - 3\bar{z} \right) dz = \oint_C \frac{e^z}{z+3} dz - 3 \oint_C \bar{z} dz.$$

By Theorem 18.4, $\oint_C \frac{e^z}{z+3} dz = 0$. However, since $\bar{z}$ is not analytic,

$$\oint_C \bar{z} dz = \int_0^{2\pi} e^{-it} (ie^{it} dt) = 2\pi i.$$

Thus

$$\oint_C \left(\frac{e^z}{z+3} - 3\bar{z} \right) dz = 0 - 3(2\pi i) = -6\pi i.$$

24. Write
$$\oint_C (z^2 + z + \operatorname{Re}(z)) dz = \oint_C (z^2 + z) dz + \oint_C \operatorname{Re}(z) dz.$$

By Theorem 18.4, $\oint_C (z^2 + z) dz = 0$. However, since $\operatorname{Re}(z) = x$ is not analytic,

$$\oint_C x dz = \oint_{C_1} x dz + \oint_{C_2} x dz + \oint_{C_3} x dz$$

where C_1 is $y = 0$, $0 \leq x \leq 1$, C_2 is $x = 1$, $0 \leq y \leq 2$, and C_3 is $y = 2x$, $0 \leq x \leq 1$. Thus,

$$\oint_C x dz = \int_0^1 x dx + i \int_0^2 dy + (1+2i) \int_1^0 x dx = \frac{1}{2} + 2i - \frac{1}{2}(1+2i) = i.$$

EXERCISES 18.3

Independence of Path

1. (a) Choosing $x = 0$, $-1 \leq y \leq 1$ we have $z = iy$, $dz = i dy$. Thus

$$\int_C (4z - 1) dz = i \int_{-1}^1 (4iy - 1) dy = -2i.$$

(b)
$$\int_C (4z - 1) dz = \int_{-i}^i (4z - 1) dz = 2z^2 - z \Big|_{-i}^i = -2i$$

2. (a) Choosing the line $y = \frac{1}{3}x$, $0 \leq x \leq 3$ we have $z = x + \frac{1}{3}xi$, $dz = (1 + \frac{1}{3}i) dx$. Thus

$$\int_C e^z dz = \int_0^3 e^{(1+\frac{1}{3}i)x} \left(1 + \frac{1}{3}i\right) dx = e^{(1+\frac{1}{3}i)x} \Big|_0^3 = e^{3+i} - e^0 = (e^3 \cos 1 - 1) + ie^3 \sin 1.$$

(b) $\int_C e^z dz = \int_0^{3+i} e^z dz = e^z \Big|_0^{3+i} = e^{3+i} - e^0 = (e^3 \cos 1 - 1) + ie^3 \sin 1$

3. The given integral is independent of the path. Thus

$$\int_C 2z dz = \int_{-2+7i}^{2-i} 2z dz = z^2 \Big|_{-2+7i}^{2-i} = 48 + 24i.$$

4. The given integral is independent of the path. Thus

$$\int_C 6z^2 dz = \int_2^{2-i} 6z^2 dz = z^3 \Big|_2^{2-i} = -15 - 24i.$$

5. $\int_0^{3+i} z^2 dz = \frac{1}{3} z^3 \Big|_0^{3+i} = 6 + \frac{26}{3}i$

6. $\int_{-2i}^1 (3z^2 - 4z + 5i) dz = z^3 - 2z^2 + 5iz \Big|_{-2i}^1 = -19 - 3i$

7. $\int_{1-i}^{1+i} z^3 dz = \frac{1}{4} z^4 \Big|_{1-i}^{1+i} = 0$

8. $\int_{-3i}^{2i} (z^3 - z) dz = \frac{1}{4} z^4 - \frac{1}{2} z^2 \Big|_{-3i}^{2i} = \frac{123}{4}$

9. $\int_{-i/2}^{1-i} (2z+1)^2 dz = \frac{1}{6} (2z+1)^3 \Big|_{-i/2}^{1-i} = -\frac{7}{6} - \frac{22}{3}i$

10. $\int_1^i (iz+1)^3 dz = \frac{1}{4i} (iz+1)^4 \Big|_1^i = -i$

11. $\int_{i/2}^i e^{\pi z} dz = \frac{1}{\pi} e^{\pi z} \Big|_{i/2}^i = -\frac{1}{\pi} - \frac{1}{\pi}i$

12. $\int_{1-i}^{1+2i} ze^{z^2} dz = \frac{1}{2} e^{z^2} \Big|_{1-i}^{1+2i} = \frac{1}{2} [e^{-3+4i} - e^{-2i}] = \frac{1}{2} (e^{-3} \cos 4 - \cos 2) + \frac{1}{2} (e^{-3} \sin 4 + \sin 2)i = 0.1918 + 0.4358i$

13. $\int_{\pi}^{\pi+2i} \sin \frac{z}{2} dz = -2 \cos \frac{z}{2} \Big|_{\pi}^{\pi+2i} = -2 \left[\cos \left(\frac{\pi}{2} + i \right) - \cos \frac{\pi}{2} \right] = 2i \sin \frac{\pi}{2} \sinh 1 = 2.3504i$

14. $\int_{1-2i}^{\pi i} \cos z dz = \sin z \Big|_{1-2i}^{\pi i} = \sin \pi i - \sin(1-2i) = i \sinh \pi - [\sinh 1 \cosh 2 - i \cos 1 \sinh 2]$
 $= -\sin 1 \cosh 2 + i(\sinh \pi + \cos 1 \sinh 2) = -3.1658 + 13.5083i$

15. $\int_{\pi i}^{2\pi i} \cosh z dz = \sinh z \Big|_{\pi i}^{2\pi i} = \sinh 2\pi i - \sinh \pi i = i \sin 2\pi - i \sin \pi = 0$

16. $\int_i^{1+\frac{\pi}{2}i} \sinh 3z dz = \frac{1}{3} \cosh 3z \Big|_i^{1+\frac{\pi}{2}i} = \frac{1}{3} \left[\cosh \left(3 + \frac{3\pi}{2}i \right) - \cosh 3i \right]$
 $= \frac{1}{3} \left[\cosh 3 \cos \frac{3\pi}{2} + i \sinh 3 \sin \frac{3\pi}{2} - \cosh 3 \right] = -\frac{1}{3} \cosh 3 - \frac{1}{3}i \sinh 3 = 0.3300 - 3.3393i$

18.3 Independence of Path

17. $\int_{-4i}^{4i} \frac{1}{z} dz = \text{Ln} z \Big|_{-4i}^{4i} = \text{Ln} 4i - \text{Ln}(-4i) = \log_e 4 + \frac{\pi}{2}i - \left(\log_e 4 - \frac{\pi}{2}i\right) = \pi i$
18. $\int_{1+i}^{4+4i} \frac{1}{z} dz = \text{Ln} z \Big|_{1+i}^{4+4i} = \text{Ln}(4+4i) - \text{Ln}(1+i) = \log_e 4\sqrt{2} + \frac{\pi}{4}i - \left(\log_e \sqrt{2} + \frac{\pi}{4}i\right) = \log_e 4 = 1.3863$
19. $\int_{-4i}^{4i} \frac{1}{z^2} dz = -\frac{1}{z} \Big|_{-4i}^{4i} = -\left[\frac{1}{4i} - \left(-\frac{1}{4i}\right)\right] = \frac{1}{2}i$
20. $\int_{1-i}^{1+\sqrt{3}i} \left(\frac{1}{z} + \frac{1}{z^2}\right) dz = \text{Ln} z - \frac{1}{z} \Big|_{1-i}^{1+\sqrt{3}i} = \log_e 2 + \frac{\pi}{3}i - \frac{1}{1+\sqrt{3}i} - \left(\log_e \sqrt{2} - \frac{\pi}{4}i - \frac{1}{1-i}\right)$
 $= \log_e \sqrt{2} + \frac{1}{4} + i \left(\frac{7\pi}{12} + \frac{\sqrt{3}}{4} + \frac{1}{2}\right) = 0.5966 + 2.7656i$

21. Integration by parts gives

$$\int e^z \cos z dz = \frac{1}{2} e^z (\cos z + \sin z) + C$$

and so

$$\begin{aligned} \int_{\pi}^i e^z \cos z dz &= \frac{1}{2} e^z (\cos z + \sin z) \Big|_{\pi}^i = \frac{1}{2} [e^i (\cos i + \sin i) - e^{\pi} (\cos \pi + \sin \pi)] \\ &= \frac{1}{2} [(\cos 1 \cosh 1 - \sin 1 \sinh 1 + e^{\pi}) + i(\cos 1 \sinh 1 + \sin 1 \cosh 1)] = 11.4928 + 0.9667i. \end{aligned}$$

22. Integration by parts gives

$$\int z \sin z dz = -z \cos z + \sin z + C$$

and so

$$\int_0^i z \sin z dz = -z \cos z + \sin z \Big|_0^i = -i \cos i + \sin i = -i \cosh 1 + i \sinh 1 = -0.3679i.$$

23. Integration by parts gives

$$\int z e^z dz = z e^z - e^z + C$$

and so

$$\int_i^{1+i} z e^z dz = e^z (z-1) \Big|_i^{1+i} = i e^{1+i} + e^i (1-i) = (\cos 1 + \sin 1 - e \sin 1) + i(\sin 1 - \cos 1 + e \cos 1) = -0.9056 + 1.7699i.$$

24. Integration by parts gives

$$\int z^2 e^z dz = z^2 e^z - 2z e^z + 2e^z + C$$

and so

$$\int_0^{\pi i} z^2 e^z dz = e^z (z^2 - 2z + 2) \Big|_0^{\pi i} = e^{\pi i} (-\pi^2 - 2\pi i + 2) - 2 = \pi^2 - 4 + 2\pi i.$$

EXERCISES 18.4

Cauchy's Integral Formulas

1. By Theorem 18.9, with $f(z) = 4$,

$$\oint_C \frac{4}{z - 3i} dz = 2\pi i \cdot 4 = 8\pi i.$$

2. By Theorem 18.10 with $f(z) = z^2$ and $f'(z) = 2z$,

$$\oint_C \frac{z^2}{(z - 3i)^2} dz = \frac{2\pi i}{1!} 2(3i) = -12\pi.$$

3. By Theorem 18.9 with $f(z) = e^z$,

$$\oint_C \frac{e^z}{z - \pi i} dz = 2\pi i e^{\pi i} = -2\pi i.$$

4. By Theorem 18.9 with $f(z) = 1 + 2e^z$,

$$\oint_C \frac{1 + 2e^z}{z} dz = 2\pi i(1 + 2e^0) = 6\pi i.$$

5. By Theorem 18.9 with $f(z) = z^2 - 3z + 4i$,

$$\oint_C \frac{z^2 - 3z + 4i}{z - (-2i)} dz = 2\pi i(-4 + 6i + 4i) = -\pi(20 + 8i).$$

6. By Theorem 18.9 with $f(z) = \frac{1}{3} \cos z$,

$$\oint_C \frac{\frac{1}{3} \cos z}{z - \frac{\pi}{3}} dz = 2\pi i \left(\frac{1}{3} \cos \frac{\pi}{3} \right) = \frac{\pi}{3} i.$$

7. (a) By Theorem 18.9 with $f(z) = \frac{z^2}{z + 2i}$,

$$\oint_C \frac{\frac{z^2}{z + 2i}}{z - 2i} dz = 2\pi i \left(-\frac{4}{4i} \right) = -2\pi.$$

- (b) By Theorem 18.9 with $f(z) = \frac{z^2}{z - 2i}$,

$$\oint_C \frac{\frac{z^2}{z - 2i}}{z - (-2i)} dz = 2\pi i \left(\frac{-4}{-4i} \right) = 2\pi.$$

8. (a) By Theorem 18.9 with $f(z) = \frac{z^2 + 3z + 2i}{z + 4}$,

$$\oint_C \frac{\frac{z^2 + 3z + 2i}{z + 4}}{z - 1} dz = 2\pi i \left(\frac{4 + 2i}{5} \right) = \pi \left(-\frac{4}{5} + \frac{8}{5} i \right).$$

18.4 Cauchy's Integral Formulas

(b) By Theorem 18.9 with $f(z) = \frac{z^2 + 3z + 2i}{z - 1}$,

$$\oint_C \frac{\frac{z^2 + 3z + 2i}{z - 1}}{z - (-4)} dz = 2\pi i \left(\frac{4 + 2i}{-5} \right) = \pi \left(\frac{4}{5} - \frac{8}{5}i \right).$$

9. By Theorem 18.9 with $f(z) = \frac{z^2 + 4}{z - i}$,

$$\oint_C \frac{\frac{z^2 + 4}{z - i}}{z - 4i} dz = 2\pi i \left(-\frac{12}{3i} \right) = -8\pi.$$

10. By Theorem 18.9 with $f(z) = \frac{\sin z}{z + \pi i}$,

$$\oint_C \frac{\frac{\sin z}{z + \pi i}}{z - \pi i} dz = 2\pi i \left(\frac{\sin \pi i}{2\pi i} \right) = i \sinh \pi.$$

11. By Theorem 18.10 with $f(z) = e^{z^2}$, $f'(z) = 2ze^{z^2}$, and $f''(z) = 4z^2e^{z^2} + 2e^{z^2}$,

$$\oint_C \frac{e^{z^2}}{(z - i)^3} dz = \frac{2\pi i}{2!} [-4e^{-1} + 2e^{-1}] = -2\pi e^{-1}i.$$

12. By Theorem 18.10 with $f(z) = z$, $f'(z) = 1$, $f''(z) = 0$, and $f'''(z) = 0$,

$$\oint_C \frac{z}{(z - (-i))^4} dz = \frac{2\pi i}{3!} (0) = 0.$$

13. By Theorem 18.10 with $f(z) = \cos 2z$, $f'(z) = -2\sin 2z$, $f''(z) = -4\cos 2z$, $f'''(z) = 8\sin 2z$, $f^{(4)}(z) = 16\cos 2z$,

$$\oint_C \frac{\cos 2z}{z^5} dz = \frac{2\pi i}{4!} (16\cos 0) = \frac{4\pi}{3}i.$$

14. By Theorem 18.10 with $f(z) = e^{-z}\sin z$, $f'(z) = e^{-z}\cos z - e^{-z}\sin z$, and $f''(z) = -2e^{-z}\cos z$,

$$\oint_C \frac{e^{-z}\sin z}{z^3} dz = \frac{2\pi i}{2!} (-2e^0\cos 0) = -2\pi i.$$

15. (a) By Theorem 18.9 with $f(z) = \frac{2z + 5}{z - 2}$,

$$\oint_C \frac{\frac{2z + 5}{z - 2}}{z} dz = 2\pi i \left(-\frac{5}{2} \right) = -5\pi i.$$

(b) Since the circle $|z - (-1)| = 2$ encloses only $z = 0$, the value of the integral is the same as in part (a).

(c) From Theorem 18.9 with $f(z) = \frac{2z + 5}{z}$,

$$\oint_C \frac{\frac{2z + 5}{z}}{z - 2} dz = 2\pi i \left(\frac{9}{2} \right) = 9\pi i.$$

(d) Since the circle $|z - (-2i)| = 1$ encloses neither $z = 0$ nor $z = 2$ it follows from the Cauchy-Goursat Theorem, Theorem 18.4, that

$$\oint_C \frac{2z + 5}{z(z - 2)} dz = 0.$$

16. By partial fractions,

$$\oint_C \frac{z}{(z-1)(z-2)} dz = 2 \oint_C \frac{dz}{z-2} - \oint_C \frac{dz}{z-1}.$$

(a) By the Cauchy-Goursat Theorem, Theorem 18.4,

$$\oint_C \frac{z}{(z-1)(z-2)} dz = 0.$$

(b) As in part (a), the integral is 0.

(c) By Theorem 18.4, $\oint_C \frac{dz}{z-2} = 0$ whereas by Theorem 18.9, $\oint_C \frac{dz}{z-1} = 2\pi i$. Thus

$$\oint_C \frac{z}{(z-1)(z-2)} dz = -2\pi i.$$

(d) By Theorem 18.9, $\oint_C \frac{dz}{z-1} = 2\pi i$ and $\oint_C \frac{dz}{z-2} = 2\pi i$. Thus

$$\oint_C \frac{z}{(z-1)(z-2)} dz = 2(2\pi i) - 2\pi i = 3\pi i.$$

17. (a) By Theorem 18.10 with $f(z) = \frac{z+2}{z-1-i}$ and $f'(z) = \frac{-3-i}{(z-1-i)^2}$,

$$\oint_C \frac{\frac{z+2}{z-1-i}}{z^2} dz = \frac{2\pi i}{1!} \left(\frac{-3-i}{(-1-i)^2} \right) = -\pi(3+i).$$

(b) By Theorem 18.9 with $f(z) = \frac{z+2}{z^2}$,

$$\oint_C \frac{\frac{z+2}{z^2}}{z-(1+i)} dz = 2\pi i \left(\frac{3+i}{(1+i)^2} \right) = \pi(3+i).$$

18. (a) By Theorem 18.10 with $f(z) = \frac{1}{z-4}$, $f'(z) = -\frac{1}{(z-4)^2}$, and $f''(z) = \frac{2}{(z-4)^3}$,

$$\oint_C \frac{\frac{1}{z-4}}{z^3} dz = \frac{2\pi i}{2!} \left(\frac{2}{-64} \right) = -\frac{\pi}{32} i.$$

(b) By the Cauchy-Goursat Theorem, Theorem 18.4,

$$\oint_C \frac{1}{z^3(z-4)} dz = 0.$$

19. By writing

$$\oint_C \left(\frac{e^{2iz}}{z^4} - \frac{z^4}{(z-i)^3} \right) dz = \oint_C \frac{e^{2iz}}{z^4} dz - \oint_C \frac{z^4}{(z-i)^3} dz$$

we can apply Theorem 18.10 to each integral:

$$\oint_C \frac{e^{2iz}}{z^4} dz = \frac{2\pi i}{3!} (-8i) = \frac{8\pi}{3}, \quad \oint_C \frac{z^4}{(z-i)^3} dz = \frac{2\pi i}{2!} (-12) = -12\pi i.$$

Thus

$$\oint_C \left(\frac{e^{2iz}}{z^4} - \frac{z^4}{(z-i)^3} \right) dz = \pi \left(\frac{8}{3} + 12i \right).$$

20. By writing

$$\oint_C \left(\frac{\cosh z}{(z-\pi)^3} - \frac{\sin^2 z}{(2z-\pi)^3} \right) dz = \oint_C \frac{\cosh z}{(z-\pi)^3} dz - \oint_C \frac{\frac{1}{8} \sin^2 z}{(z-\frac{\pi}{2})^3} dz$$

18.4 Cauchy's Integral Formulas

we apply Theorem 18.4 to the first integral and Theorem 18.10 to the second:

$$\oint_C \frac{\cosh z}{(z-\pi)^3} dz = 0, \quad \oint_C \frac{\frac{1}{8} \sin^2 z}{(z-\frac{\pi}{2})^3} dz = \frac{2\pi i}{2!} \left(-\frac{1}{4} \sin^2 \frac{\pi}{2} \right) = -\frac{\pi}{4} i.$$

Thus

$$\oint_C \left(\frac{\cosh z}{(z-\pi)^3} - \frac{\sin^2 z}{(2z-\pi)^3} \right) dz = \frac{\pi}{4} i.$$

21. We have

$$\oint_C \frac{1}{z^3(z-1)^2} dz = \oint_{C_1} \frac{\frac{1}{(z-1)^2}}{z^3} dz + \oint_{C_2} \frac{\frac{1}{z^3}}{(z-1)^2} dz$$

where C_1 and C_2 are the circles $|z| = 1/3$ and $|z-1| = 1/3$, respectively. By Theorem 18.10,

$$\oint_{C_1} \frac{\frac{1}{(z-1)^2}}{z^3} dz = \frac{2\pi i}{2!} (6) = 6\pi i, \quad \oint_{C_2} \frac{\frac{1}{z^3}}{(z-1)^2} dz = \frac{2\pi i}{1!} (-3) = -6\pi i.$$

Thus

$$\oint_C \frac{1}{z^3(z-1)^2} dz = 6\pi i - 6\pi i = 0.$$

22. We have

$$\oint_C \frac{1}{z^2(z^2+1)} dz = \oint_{C_1} \frac{\frac{1}{z^2(z+i)}}{z-i} dz + \oint_{C_2} \frac{\frac{1}{z^2+1}}{z^2} dz$$

where C_1 and C_2 are the circles $|z-i| = 1/3$ and $|z| = 1/8$, respectively. By Theorems 18.9 and 18.10,

$$\oint_{C_1} \frac{\frac{1}{z^2(z+i)}}{z-i} dz = 2\pi i \left(\frac{1}{-2i} \right) = -\pi, \quad \oint_{C_2} \frac{\frac{1}{z^2+1}}{z^2} dz = \frac{2\pi i}{1!} (0) = 0.$$

Thus

$$\oint_C \frac{1}{z^2(z^2+1)} dz = -\pi.$$

23. We have

$$\oint_C \frac{3z+1}{z(z-2)^2} dz = \oint_{C_1} \frac{\frac{3z+1}{z}}{(z-2)^2} dz - \oint_{C_2} \frac{\frac{3z+1}{(z-2)^2}}{z} dz$$

where C_1 and C_2 are the closed portions of the curve C enclosing $z = 2$ and $z = 0$, respectively. By Theorems 18.10 and 18.9,

$$\oint_{C_1} \frac{\frac{3z+1}{z}}{(z-2)^2} dz = \frac{2\pi i}{1!} \left(-\frac{1}{4} \right) = -\frac{\pi}{2} i, \quad \oint_{C_2} \frac{\frac{3z+1}{(z-2)^2}}{z} dz = 2\pi i \left(\frac{1}{4} \right) = \frac{\pi}{2} i.$$

Thus

$$\oint_C \frac{3z+1}{z(z-2)^2} dz = -\frac{\pi}{2} i - \frac{\pi}{2} i = -\pi i.$$

24. We have

$$\oint_C \frac{e^{iz}}{(z^2+1)^2} dz = \oint_{C_1} \frac{\frac{e^{iz}}{(z+i)^2}}{(z-i)^2} dz - \oint_{C_2} \frac{\frac{e^{iz}}{(z-i)^2}}{(z-(-i))^2} dz$$

where C_1 and C_2 are the closed portions of the curve C enclosing $z = i$ and $z = -i$, respectively. By Theorem 18.10,

$$\oint_{C_1} \frac{e^{iz}}{(z+i)^2} dz = \frac{2\pi i}{1!} \left(\frac{-4e^{-1}}{-8i} \right) = \pi e^{-1}, \quad \oint_{C_2} \frac{e^{iz}}{(z-i)^2} dz = \frac{2\pi i}{1!} \left(\frac{0}{8i} \right) = 0.$$

Thus

$$\oint_C \frac{e^{iz}}{(z^2+1)^2} dz = \pi e^{-1}.$$

CHAPTER 18 REVIEW EXERCISES

1. True
2. False
3. True
4. True
5. 0
6. $\pi(-16 + 8i)$
7. $\pi(6\pi - i)$
8. a constant function
9. True (Use partial fractions and write the given integral as two integrals.)
10. True
11. integer not equal to -1 ; -1
12. 12π
13. Since $f(z) = z$ is entire, $\int_C (x + iy) dz$ is independent of the path C . Thus

$$\oint_C (x + iy) dz = \int_{-4}^3 z dz = \frac{z^2}{2} \Big|_{-4}^3 = -\frac{7}{2}.$$

14. We have $\int_C (x - iy) dz = \int_{C_1} (x - iy) dz + \int_{C_2} (x - iy) dz + \int_{C_3} (x - iy) dz$

On C_1 , $x = 4$, $0 \leq y \leq 2$, $z = 4 + iy$, $dz = i dy$,

$$\int_{C_1} (4 - iy)i dy = i \int_0^2 (4 - iy) dy = i \left(4y - \frac{i}{2}y^2 \right) \Big|_0^2 = 2 + 8i.$$

On C_2 , $y = 2$, $-4 \leq x \leq 3$, $z = x + 2i$, $dz = dx$,

$$\int_{C_2} (x - 2i) dx = \int_{-4}^3 (x - 2i) dx = \frac{1}{2}x^2 - 2ix \Big|_{-4}^3 = -\frac{7}{2} - 14i.$$

On C_3 , $x = 3$, $0 \leq y \leq 2$, $z = 3 + iy$, $dz = i dy$,

$$\int_{C_3} (3 - iy)i dy = i \int_0^2 (3 - iy) dy = i \left(3y - \frac{i}{2}y^2 \right) \Big|_0^2 = -2 - 6i.$$

Thus

$$\int_C (x - iy) dz = 2 + 8i - \frac{7}{2} - 14i - 2 - 6i = -\frac{7}{2} - 12i.$$

CHAPTER 18 REVIEW EXERCISES

$$15. \int_C |z^2| dz = \int_0^2 (t^4 + t^2) dt + 2i \int_0^2 (t^5 + t^3) dt = \frac{136}{15} + \frac{88}{3}i$$

$$16. \int_C e^{\pi z} dz = \frac{1}{\pi} \int_i^{1+i} e^{\pi z} (\pi dz) = \frac{1}{\pi} e^{\pi z} \Big|_i^{1+i} = \frac{1}{\pi} (1 - e^{\pi})$$

$$17. \text{ By the Cauchy-Goursat Theorem, Theorem 18.4, } \oint_C e^{\pi z} dz = 0.$$

$$18. \int_{3i}^{1-i} (4z - 6) dz = 2z^2 - 6z \Big|_{3i}^{1-i} = 12 + 20i$$

$$19. \int_C \sin z dz = \int_1^{1+4i} \sin z dz = -\cos z \Big|_1^{1+4i} = \cos 1 - \cos(1 + 4i) = -14.2144 + 22.9637i$$

$$20. \int_C (4z^3 + 3z^2 + 2z + 1) dz = \int_0^{2i} (4z^3 + 3z^2 + 2z + 1) dz = z^4 + z^3 + z^2 + z \Big|_0^{2i} = 12 - 6i$$

$$21. \text{ On } |z| = 1, \text{ let } z = e^{it}, dz = ie^{it} dt, \text{ so that}$$

$$\oint_C (z^{-2} + z^{-1} + z + z^2) dz = i \int_0^{2\pi} (e^{-2it} + e^{-it} + e^{it} + e^{2it}) e^{it} dt = -e^{-it} + it + \frac{1}{2}e^{2it} + \frac{1}{3}e^{3it} \Big|_0^{2\pi} = 2\pi i.$$

$$22. \text{ By partial fractions and Theorem 18.9,}$$

$$\oint_C \frac{3z + 4}{z^2 - 1} dz = \frac{7}{2} \oint_C \frac{1}{z - 1} dz - \frac{1}{2} \oint_C \frac{1}{z - (-1)} dz = \frac{7}{2}(2\pi i) - \frac{1}{2}(2\pi i) = 6\pi i.$$

$$23. \text{ By Theorem 18.10 with } f(z) = e^{-2z}, f'(z) = -2e^{-2z}, f''(z) = 4e^{-2z}, \text{ and } f'''(z) = -8e^{-2z},$$

$$\oint_C \frac{e^{-2z}}{z^4} dz = \frac{2\pi i}{3!} (-8) = -\frac{8\pi}{3} i.$$

$$24. \text{ By Theorem 18.10 with } f(z) = \frac{\cos z}{z - 1} \text{ and } f'(z) = \frac{\sin z - \cos z - z \sin z}{(z - 1)^2},$$

$$\oint_C \frac{\frac{\cos z}{z - 1}}{z^2} dz = \frac{2\pi i}{1!} \left(\frac{-1}{1} \right) = -2\pi i.$$

$$25. \text{ By Theorem 18.9 with } f(z) = \frac{1}{2(z + 3)},$$

$$\oint_C \frac{\frac{1}{2(z + 3)}}{(z - (-1/2))} dz = 2\pi i \left(\frac{1}{5} \right) = \frac{2\pi}{5} i.$$

$$26. \text{ Since the function } f(z) = z/\sin z \text{ is analytic within and on the given simple closed contour } C, \text{ it follows from the Cauchy-Goursat Theorem, Theorem 18.4, that}$$

$$\oint_C z \csc z dz = 0.$$

$$27. \text{ Using the principle of deformation of contours we choose } C \text{ to be the more convenient circular contour } |z + i| = \frac{1}{4}. \text{ On this circle } z = -i + \frac{1}{4}e^{it} \text{ and } dz = \frac{1}{4}ie^{it} dt. \text{ Thus}$$

$$\oint_C \frac{z}{z + i} dz = i \int_0^{2\pi} \left(\frac{1}{4}e^{it} - i \right) dt = 2\pi.$$

28. (a) By Theorem 18.9 with $f(z) = \frac{e^{i\pi z}}{2(z-2)}$,

$$\oint_C \frac{\overline{e^{i\pi z}}}{2(z-2)} dz = 2\pi i \left(\frac{e^{i\pi/2}}{-3} \right) = \frac{2\pi}{3}.$$

(b) By Theorem 18.9 with $f(z) = \frac{e^{i\pi z}}{2z-1}$,

$$\oint_C \frac{\overline{e^{i\pi z}}}{2z-1} dz = 2\pi i \left(\frac{e^{2\pi i}}{3} \right) = \frac{2\pi}{3} i.$$

(c) By the Cauchy-Goursat Theorem, Theorem 18.4,

$$\oint_C \frac{e^{i\pi z}}{2z^2 - 5z + 2} dz = 0.$$

29. For $f(z) = z^n g(z)$ we have $f'(z) = z^n g'(z) + n z^{n-1} g(z)$ and so

$$\frac{f'(z)}{f(z)} = \frac{z^n g'(z) + n z^{n-1} g(z)}{z^n g(z)} = \frac{g'(z)}{g(z)} + \frac{n}{z}.$$

Thus by Theorem 18.4 and (4) of Section 18.2,

$$\oint_C \frac{f'(z)}{f(z)} dz = \oint_C \frac{g'(z)}{g(z)} dz + n \oint_C \frac{1}{z} dz = 0 + n(2\pi i) = 2n\pi i.$$

30. We have

$$\left| \int_C \operatorname{Ln}(z+1) dz \right| \leq |\max \text{ of } \operatorname{Ln}(z+1) \text{ on } C| \cdot 2,$$

where 2 is the length of the line segment. Now

$$|\operatorname{Ln}(z+1)| \leq |\log_e(z+1)| + |\operatorname{Arg}(z+1)|.$$

But $\max \operatorname{Arg}(z+1) = \pi/4$ when $z = i$ and $\max |z+1| = \sqrt{10}$ when $z = 2+i$. Thus,

$$\left| \int_C \operatorname{Ln}(z+1) dz \right| \leq \left(\frac{1}{2} \log_e 10 + \frac{\pi}{4} \right) 2 = \log_e 10 + \frac{\pi}{2}.$$